

Generative and Predictive Models of Action Potentials in Venus Fly Traps

C: Do Plants Dream of Electric Bees?

By: Will Brooks, Munish Nidadavolu, Maggie Samuta, Nathan Glymph, and Isaac Paolino

Abstract:

Some plants, such as the Venus Fly Trap (*Dionaea muscipula*), utilize electrical signals resembling animal action potentials (APs) to trigger rapid mechanical responses for prey capture. This project aimed to determine if these electrical responses can be reliably measured and modeled to predict AP generation, timing, trap closure and the release of digestive enzymes. Using a Plant SpikerBox and MATLAB for signal processing, APs were recorded from five traps under varying stimulus conditions. A custom low-pass FIR filter (12 Hz cutoff) and normalized cross-correlation (NCC) filters were developed to detect AP events, trap closure, and digestion-related activity. Two primary models were created: an event detection model and a generative model that predicts electrophysiological output based on stimulus count and inter-stimulus interval (ISI). Results indicated that the event detection model was highly effective, achieving an overall F1-score of 0.99 with no false negatives across 24 trials. The average latency from stimulus to AP onset was 23 ms, and the recovery time required for a second AP to trigger closure was found to be 22 seconds. While the generative model accurately captured temporal dynamics and waveform shapes, it showed variability in reproducing precise voltage magnitudes. These findings demonstrate that plant electrophysiological activity can be reliably modeled, offering potential applications in precision agriculture and plant neurobiology. Results from this experiment can be used to form a CNN and RNN based model to process venus fly trap electrophysiology (EP) regardless of amplitude or timing of the signal.

Problem Statement and Relevance:

Although plants lack neurons and a centralized nervous system, they still generate electrical signals that closely resemble action potentials observed in animal systems [1]. These signals play a critical role in processes such as environmental sensing and rapid mechanical responses, as seen in *Dionaea muscipula*, more commonly known as the Venus Fly Trap. However, unlike animal electrophysiology, plant electrical signaling is less well understood, particularly in terms of how stimulus patterns influence action potential timing, frequency, and downstream physiological responses.

The primary problem addressed in this project is whether the electrical responses of the Venus Fly Trap can be reliably measured and modeled to predict action potential generation, timing, and recovery dynamics under controlled stimulation conditions. Specifically, we aim to determine if features such as inter-stimulus interval (ISI), stimulus count, and latency can be used to construct a productive model that accurately reflects the plant's electrophysiological behavior and its decision to close or maintain an open trap.

This problem is relevant for several reasons. First, it provides insight into how non-neural biological systems encode and process information using electrical signals, expanding our understanding of electrophysiology beyond traditional animal models. Second, developing a predictive framework for plant action potentials offers a low-cost and accessible platform for studying signal processing, system identification, and bioelectric phenomena. Finally, this work has broader implications for bio-inspired sensing systems and the development of novel

interfaces that use biological electrical signals, particularly in contexts where simple, energy-efficient detection mechanisms are advantageous.

Background:

For this project, we decided to see how action potentials behave in plants. Specifically, we observed action potential behavior in *Dionaea muscipula* (Venus Fly Trap). Action potentials are rapid changes in the potential of a cell's membrane, which is determined by the concentration of various ions inside and outside of the cell [2]. In humans, these action potentials are transported along nerve cells and can be used to encode signal strength. However, things are a little different in plants. Plants do not have the same neuronal anatomy as humans and other animals that are associated with action potentials and neuronal activity. This is because plants do not have brains or neurons [1]. Despite this fact, plants still produce action potentials similar to those in humans. For example, Venus Fly Traps use action potentials to trigger trap closing and convey stimulation information [3]. This example is useful in that the result of the action potential is readily viewable, which is why they are a good candidate for the plant model used in this experiment.

Venus Fly Traps are one of the few carnivorous plants. They are not as efficient and active with photosynthesis like standard plants, which is why they need to use their traps to capture and digest insects for nutrients. Their “traps” are really just special leaves that evolved over time [4]. So how do these traps work? Located inside the trap are special trigger hairs. When these trigger hairs are disturbed, they activate mechanoreceptors which open ion channels, which then induce action potentials if enough receptors are activated [5]. Venus fly traps then count the action potentials, and different responses happen depending on the number of action potentials. With one action potential, nothing happens. This is likely because one stimulation could be an insect, but it could also be pressure from wind or another factor. With a second action potential quickly following the first, within around 15-20 seconds, the trap begins to close. This is because it is more likely that the stimulations are from an insect rather than other factors. Upon a third action potential, the plant further responds in the form of releasing digestive enzymes to begin to break down the insect. With four or more action potentials, caused by the insect constantly moving and struggling within the trap, the plant kicks into the full digestion process [6]. A benefit of this interesting response to the number of action potentials is that we were able to record the action potentials as they came and visualize the response based on the number of action potentials triggered.

Methods:

Setup:

As shown in Figure 1, a venus fly trap was attached to two recording electrodes. The positive lead was attached to the side/spine of the trap using electrode gel to reduce noise. The ground electrode was placed in the soil of the plant. The electrodes feed into a Plant SpikerBox which amplifies the voltage between the two electrodes and sends the signal to a computer running the Spike Recorder software.

Recording:

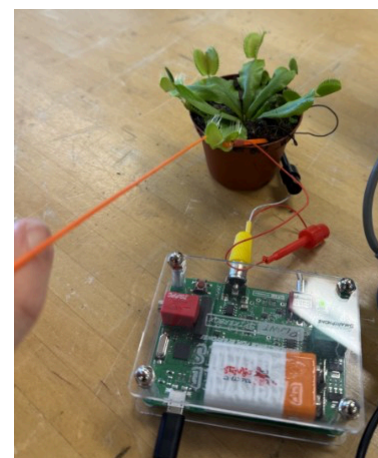


Figure 1. Experimental Setup.
Positive lead shown in red,
ground shown in black. Poking
rod shown in orange.

Spike Recorder takes the signal and saves it to a .wav file for future processing. We elected to create our own filter instead of using the built in low pass filter. To record action potentials, we recorded the response both on a slow motion camera (240 fps) and the Spike Recorder app. To stimulate the trap, we used a plastic rod and rubbed the interior hairs of the trap. Simultaneously, the person stimulating would lift their thumb upon touching the hair. Latency can then be calculated by subtracting the start of the AP and the start of the thumb lift, determined by the slow motion video. Crucially, all components of the set up, including the computer running Spike Recorder must be in the camera frame.

Stimuli Patterns Explored:

To explore the electrophysiology of the venus fly trap, the following combinations were explored:

1. Single stimuli spaced > 40 seconds apart to generate model shapes and latency
2. 2 stimuli applied starting at 40 seconds apart and decreasing inter-stimulus interval (ISI) by two seconds until the trap closed. Recording continued to determine the cause of trap closure
3. 2 stimuli applied 0.5 seconds apart to determine if there is a refractory period before trap closure
4. The trap was stimulated after closure once to determine the effect of a live insect moving in the trap. This position of the experiment was only explored once to preserve trap integrity for other recordings.

In total, there were 11 one tap stimuli, 22 two tap leaving the trap open, and 1 two tap stimulus that closed the trap with an additional third tap to see digestion spiking. Testing was performed across two plants and five traps.

If a trap did not produce action potentials despite stimulation, it was presumed dead and a new trap was used. Experiments were conducted across two venus fly trap plants and four traps total.

Signal Processing:

The .wav files were brought into MATLAB R2025b for signal processing. First the signals were put through a low pass FIR filter with order 50 and transition region of 5-10 Hz. FIR was used to prevent phase distortion because a tight transition region was unnecessary. The cutoff was determined through power spectral analysis. Individual AP signals were then aligned by the start of their first action potential.

To detect an action potential event, a correlation filter was established by calculating the average AP response and isolating that waveform. This form was passed over the signal, generating spikes at every action potential. This can be thresholded to determine the middle of every AP and the duration of the filter can be used to find the start of the AP.

Latency was calculated by taking the time elapsed between triggering the trap hairs and the start of the AP. This was determined using a slow motion camera.

Model Generation and Evaluation:

We created two models:

1. A prediction model that takes the number of stimuli and ISI and predicts the EP output and whether the trap stays open or closes. This uses latency data, ISI, stimulation count, and recovery dynamics evaluated signal recording. The model prediction is then overlaid on the empirical data plot for evaluation.
2. An event detection model that can determine how many APs occurred and whether the trap closes based on the signal recorded. This applies correlation filters to determine event timing. Events are overlaid on the empirical EP plot and verified by inspection.

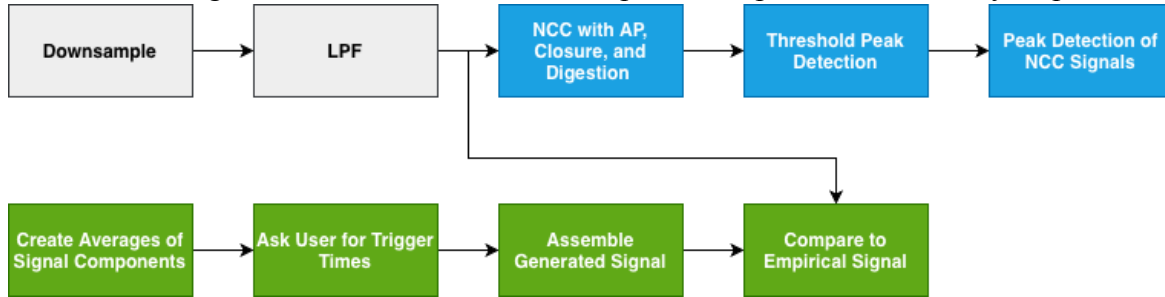


Figure 2: Signal Processing Pipeline. Event detection and generative model processing path, with preprocessing in gray, the generative model in green, and event detection in blue.

The event detection model in Figure 2 leverages normalized cross correlation of the signal with zero-mean correlation filters representing each event to determine event occurrence. This is then thresholded based on the percent match of the correlation filter and the signal. The peaks are found above the threshold, corresponding to an action potential. The generative model creates average signals of the various components, asks the user what times they want to trigger the hair, and assembles a generated signal. The empirical signals can be compared to the generated signal.

Results:

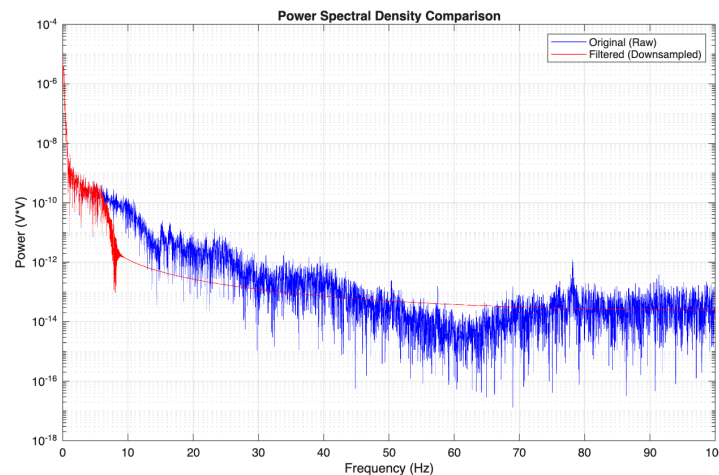


Figure 3: Power Spectra Density (PSD) Before (above) and After (below) Filtering. Signals were downsampled to 200 Hz to speed up processing.

The frequency content of the recorded signals before and after filtering is shown in Figure 3. In the raw signal (top), power is distributed across the full frequency range from 0 to 100 Hz, with a gradual decay in magnitude as frequency increases and noticeable variability

across higher frequencies. After filtering (bottom), power above approximately 25-30 Hz is significantly attenuated, resulting in a sharp drop in magnitude and a flattened spectral profile at higher frequencies. The filtered signal retains dominant low-frequency components while suppressing higher frequency contributions.

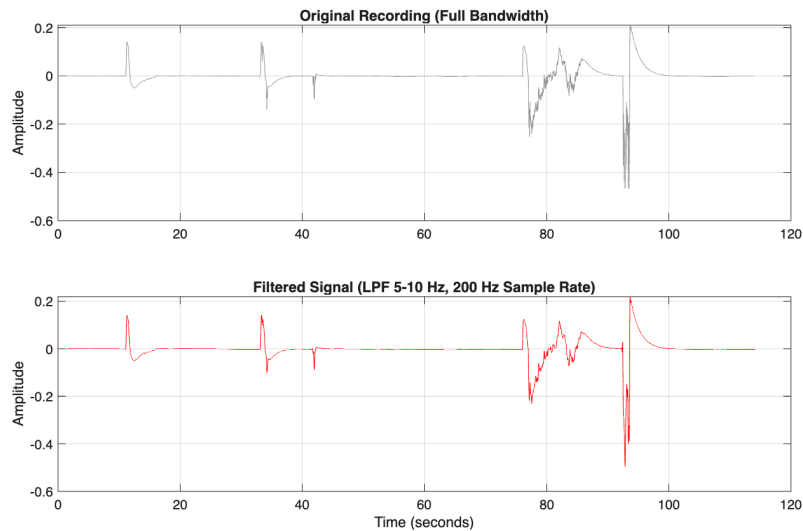


Figure 4: Signal Recordings Before (above) and After (below) the Application of a Low Pass Filter. Signals were downsampled to 200 Hz to speed up processing.

Figure 4 provides a time-domain comparison of representative recordings. The raw signal exhibits noticeable baseline drift and high-frequency fluctuations, while the filtered signal appears smoother, with reduced noise and more clearly defined voltage deflections corresponding to action potential events. Note, amplitude units are in voltage, but voltage scale was not provided by the SpikerBox software. This applies

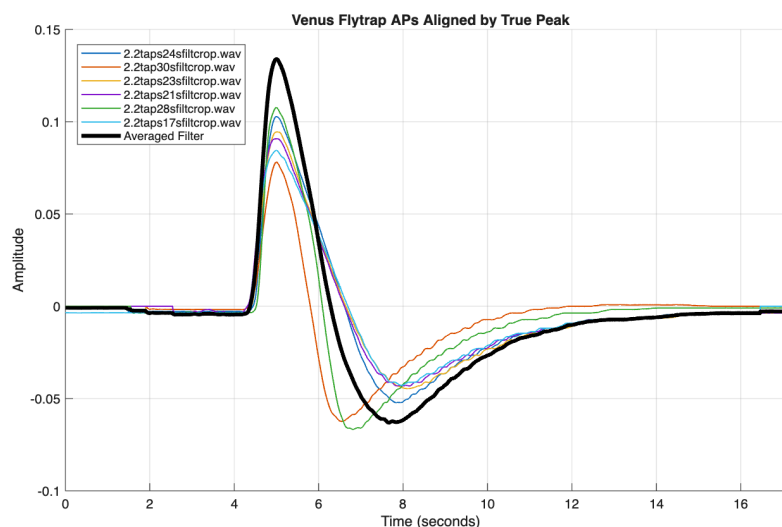


Figure 5: Action Potential Shape Determination. Various .wav files represented by different colors. Averaged filter shown in black.

The formation of the correlation filter used for event detection and generative modeling is illustrated in Figure 5. Individual action potential events were temporally aligned at their onset and averaged to generate a representative waveform template. The resulting template was used as a reference for subsequent signal processing. The action potentials including hyperpolarization lasted 22 seconds on average.

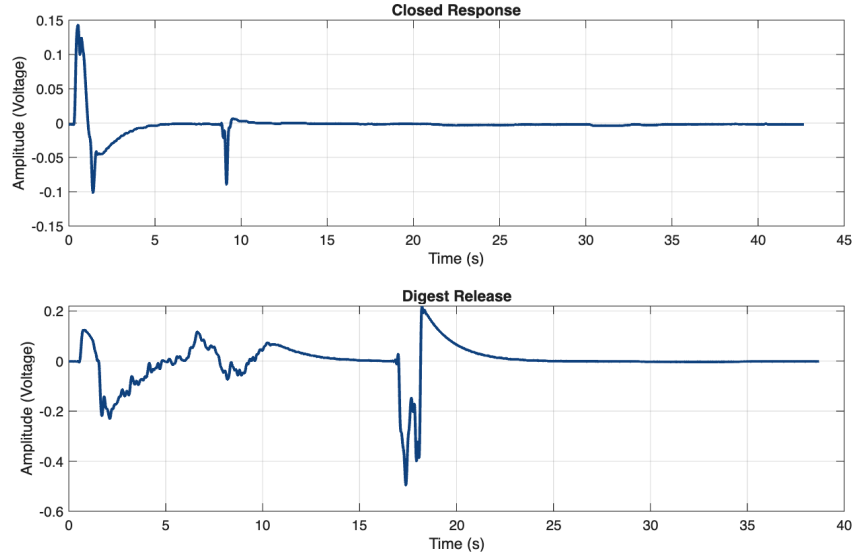


Figure 6: Action Potential Shape Determination for Trap Closure (above) and Digestion (below). .wav files of events associated with each were aligned at a common 0 and averaged for template generation.

Separate correlation filter templates corresponding to different physiological conditions are presented in Figure 6. Action potential events associated with trap closure (above) and digestion (below) were aligned and averaged to generate distinct waveform templates for each condition.

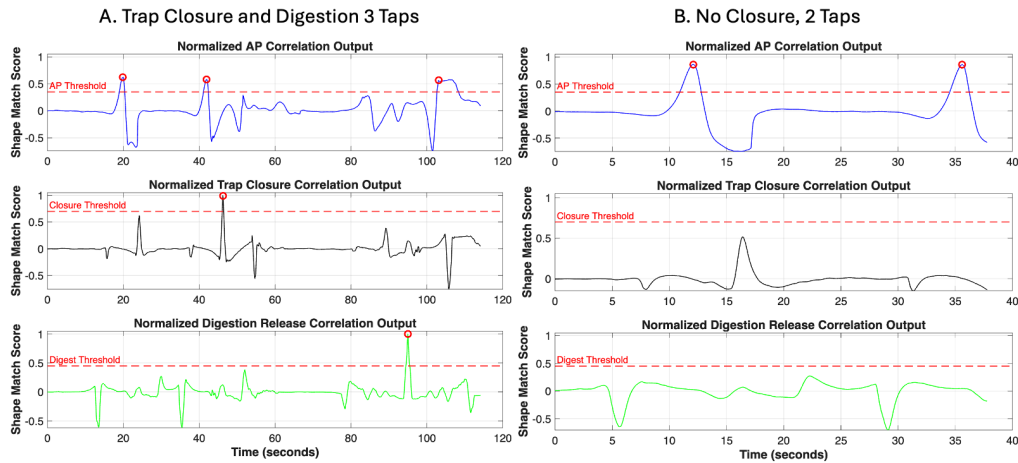


Figure 7: Application of Correlation Filters for A) 3 Taps and B) 2 Taps. Trap closure and digestion visualized after the conditions in A, but not in B.

Figure 7 illustrates the application of correlation filters under different stimulation conditions. Panel A shows the response to three stimuli, while Panel B shows the response to two stimuli. For each condition, the normalized correlation outputs for action potentials (top), trap closure (middle), and digestion release (bottom) are displayed along with their respective detection thresholds. In Panel A, multiple peaks in the action potential correlation output exceed the defined threshold, indicating detection of the three successive taps. Corresponding peaks are also observed in the trap closure and digestion correlation outputs, with threshold crossings visible in both signals. In contrast, Panel B shows two distinct peaks in the action potential correlation output that exceed the threshold, corresponding to two detected stimuli. The trap closure and digestion correlation outputs in this case remain below their respective thresholds, with no threshold crossings observed.

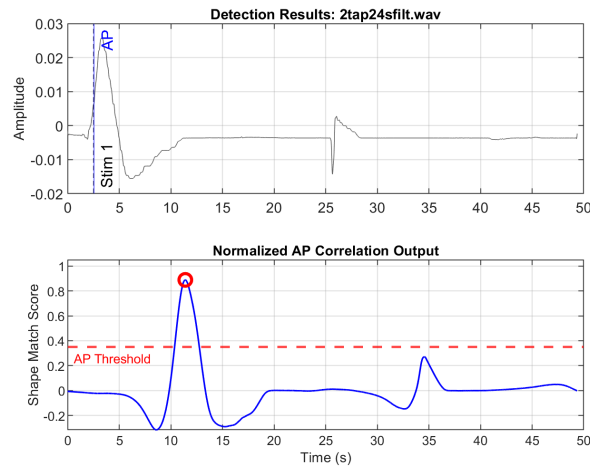


Figure 8: *Reduced Electrical Response and Failed Action Potential Detection in Deteriorating Trap.*

An example of the reduced electrical responsiveness in a deteriorating trap is shown in Figure 8. Although a stimulus is applied, the recorded signal exhibits only small-amplitude fluctuations, without a distinct AP waveform. In the normalized correlation output, no peaks exceed the defined detection threshold, indicating the absence of detected action potential events in deteriorating traps. Data from the dying traps were not included in our metrics.

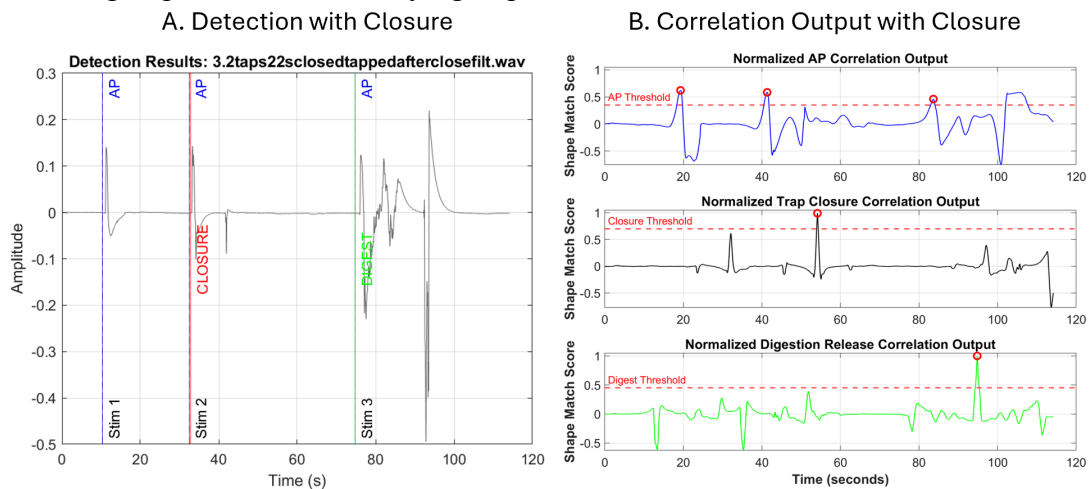


Figure 9: Trap Closure A) Detection Results and B) Correlation Outputs. Time of stimulation indicated along with their corresponding action potentials. Thresholds for action potential generation, trap closure, and digestion indicated in red on B.

Figure 9 shows the detection results (left) of the trap closure after 3 stimuli next to the correlation outputs described in Figure 7A. The peaks in Panel B exceeded their thresholds for action potentials, trap closure, and digestion, which resulted in the closure and beginning of digestion in the Venus Fly Plants. The visualization of their action potentials and their relationship to the time of stimulation are shown in Panel A. The digestion downward spike occurs 3.36 seconds after the end of hyperpolarization. Digestion start is characterized by a smaller amplitude AP starting 1.00 seconds after hyperpolarization of the first AP of digestion. Then a jagged hyperpolarization spike happens 11 seconds after the initial hyperpolarization before depolarizing and returning to zero.

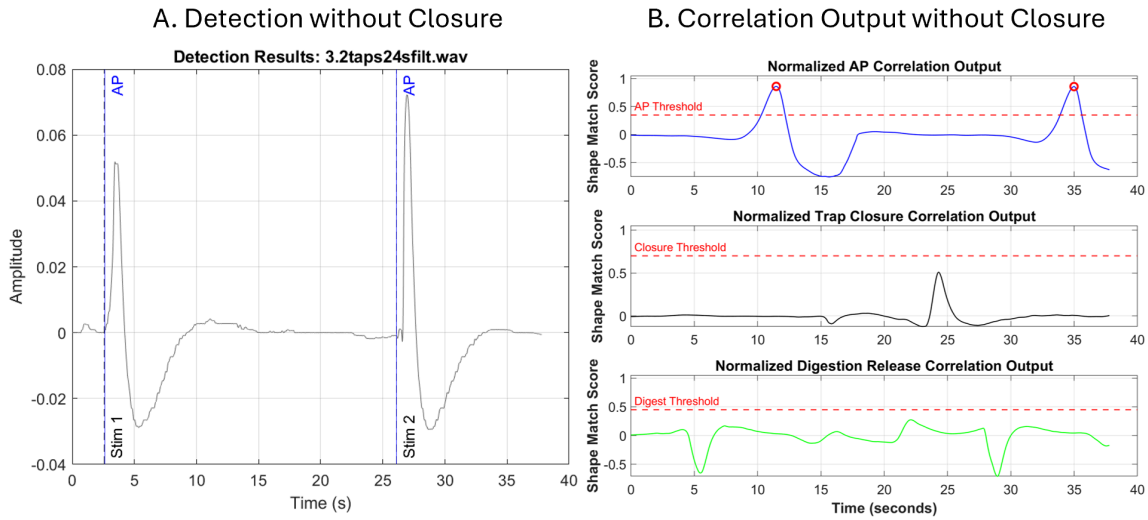


Figure 10: No Trap Closure A) Detection Results and B) Correlation Outputs. Time of stimulation indicated along with their corresponding action potentials. Thresholds for action potential generation, trap closure, and digestion indicated in red on B.

Figure 10 visualizes the detection results (left) of the trap closure after 3 stimuli next to the correlation outputs described in Figure 7B. The peaks in Panel B failed to exceed their thresholds for trap closure and digestion, and neither occurred in the Venus Fly Plants. They did however exceed the threshold for an action potential. The visualization of their action potentials and their relationship to the time of stimulation are shown in Panel A.

Table 1: Event Detection Model Performance Metrics

	APs	Closure	Digestion	Overall
Precision	1	0.5	1	0.98
Recall	1	1	1	1
F1	1	0.667	1	0.99

Table 1 summarizes the event detection performance metrics across 24 trials including those with one stimulus, two stimuli without closure, and two stimuli with closure and post-closure stimulus (digestion activity). Precision, recall, and F1-score are reported for overall event detection, as well as separate performance metrics for action potential detection, trap closure identification, and digestion-related activity. Overall the model produced no false negatives, and only one false positive closure on a trial where the plant was tapped twice 30 seconds apart, but did not close.

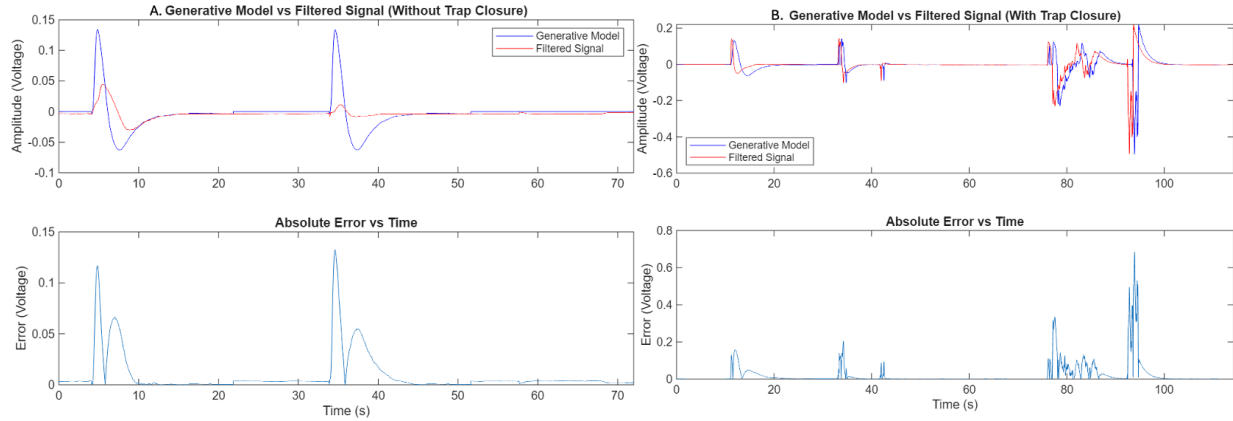


Figure 11: Generative Model Trap. A No Trap Closure and B Trap Closure: (above) Signal comparison with filtered signal given identical stimulus time compared to the model. (below) Absolute error of model vs filtered signal.

As seen in Figure 11 A, the generative model (above) shows action potentials occurring at approximately the same time as the action potentials that actually occurred when fed the same stimulation times and number of stimuli for the case of no trap closure. They noticeably differ, however, in magnitude as shown by the absolute error (below) between the signals. Similarly to Figure 11 B also shows that the generative model (above) has action potentials overlaid on the empirical data. The difference being that a trap closure did occur from these stimulations. They once again differ in magnitude as shown by the absolute error (below) between the signals.

Initial stimulus to the start of an action potential is defined as the latency had an average of 230 ms. Amplitude varied widely across plants and electrode locations, however relative amplitude and waveform shape remained consistent across studies in Figures 6-10, with the exception of the dying trap's second AP. Recovery time, defined as the time before which a second AP will trigger trap closure was found to be 22 seconds. We were unable to stimulate the trap twice within a tight enough window to observe any refractory period.

Discussion:

From the plots and figures above, it is visible that our model did an effective job of capturing the dynamics of Venus Fly Traps. From figure 11 A, one can see that the action potential timing and mechanics that our model created closely matches the timing of the actual plant, although the amplitude is different. The filtering and modeling pipeline accurately captured the temporal dynamics and overall waveform shape of the Venus flytrap signals, but it did not reproduce their amplitude with the accuracy we were hoping for. This answers our

primary question: is it possible to reliably predict and model the electrophysiological activity of a Venus Fly Trap? Yes, it is very much possible.

Further evidence can be seen in figure 11 B. In figure 11 B, we overlaid the action potential sequence created by the Venus Fly Trap with our action potential sequence created by our model. Although the model we generated is not a one-to-one match, they are extremely similar. The model signal leads the actual signal by multiple seconds. However, the signal patterns and signal amplitudes are extremely similar.

Although our system reached a high level of accuracy, improvements could be made. One area of improvement would be utilizing machine learning during the filtering process.

Specifically, using convolution neural networks (CNN) or recurrent neural networks (RNN) to improve signal processing. In figure 2, in the third block, it says “NCC with AP”, which is the correlation filter we created and applied to the action potentials. Instead, a CNN or RNN could be utilized and it would replace the correlation filter, which would increase the accuracy of our pipeline, and in turn increase accuracy of our model.

One of the main implications of this study is in the application of robotics. The Venus Fly Trap utilizes electrical communication (via action potentials) to control the mechanical trap. The model created in this study deepens the understanding of how the signal affects trap closure. Similar electrical systems could be implemented in robots to increase efficiency. Mimicking biology is a popular strategy in engineering as biology has had millions, if not billions, of years to perfect itself. The Venus Fly Trap’s electromechanics could be utilized to create or improve robotics.

The limitations of this study are related to our access to healthy Venus Fly Traps. The traps were old and likely underfed and underwatered when we got them, and thus died quickly. Because of this, we never achieved 4 action potentials in succession, and we only got one trap to close. This limited the data we were able to use in the generation of our predictive models. For this reason, it is recommended that a more thorough experiment be done with many healthy plants to achieve more robust models.

In the future, it would be useful to test other plants that utilize electrical communication, such as those *Mimosa Pudica* and *Drosera*. Understanding the similarities and differences between the three species would be pivotal in deepening the understanding of plant neurobiology and electromechanics. This could be expanded to other plants to better implement precision agriculture and improve food supply. As mentioned earlier, implementing a machine learning based filtering system would increase accuracy and further tune the model. Lastly, seeing the exact ion channels and membrane mechanics would be ideal to see exactly how the action potentials are being influenced by ions and membranes.

References:

1. A. Draguhn, J. M. Mallatt, and D. G. Robinson, “Anesthetics and plants: no pain, no brain, and therefore no consciousness,” *Protoplasma*, vol. 258, no. 2, pp. 239–248, Sep. 2020, doi: <https://doi.org/10.1007/s00709-020-01550-9>
2. M. H. Grider, R. Jessu, and R. Kabir, “Physiology, Action Potential,” National Library of Medicine, 2023. <https://www.ncbi.nlm.nih.gov/books/NBK538143/>
3. A. G. Volkov, T. Adesina, and E. Jovanov, “Closing of venus flytrap by electrical stimulation of motor cells,” *Plant signaling & behavior*, vol. 2, no. 3, pp. 139–45, 2007, doi: <https://doi.org/10.4161/psb.2.3.4217>

4. J. Kruse et al., “Strategy of nitrogen acquisition and utilization by carnivorous *Dionaea muscipula*,” *Oecologia*, vol. 174, no. 3, pp. 839–851, Oct. 2013, doi: <https://doi.org/10.1007/s00442-013-2802-9>
5. A. G. Volkov, T. Adesina, V. S. Markin, and E. Jovanov, “Kinetics and Mechanism of *Dionaea muscipula* Trap Closing,” *Plant Physiology*, vol. 146, no. 2, pp. 694–702, Dec. 2007, doi: <https://doi.org/10.1104/pp.107.10824>
6. J. Böhm et al., “The Venus Flytrap *Dionaea muscipula* Counts Prey-Induced Action Potentials to Induce Sodium Uptake,” *Current Biology*, vol. 26, no. 3, pp. 286–295, Feb. 2016, doi: <https://doi.org/10.1016/j.cub.2015.11.057>

Gemini and Claude LLMs were used in the revision of this report and generation of code.

Appendix:

Check out the GitHub Repo here for full code. Use the final-submission branch only:

<https://github.com/ijpaolin/VFTEP/tree/final-submission>

Code documentation: Clone the entire final submission branch into MATLAB. Run the MLX file. Section 0 is the title. Section 1 imports the data and passes it through an LPF. Section 2 crops signals to be used in correlation filtering. Section 3 averages the cropped signals to form an AP correlation filter. Section 4 calculates the remaining AP shapes. Section 5 is the event detection model. You must input the name of the file you wish to process before running that section. Section 7 is the generative model. You must input the stimulus number and times while running the code. Sections 8 and beyond are all plots used in this report. Make sure all code is run from the proper directory as issued in GitHub. Also, ensure you have the `cropped_data` folder before running to avoid errors.